Module 10: Epigenetics — Study Questions

1. What does "gene expression" mean? Why don't all cells express all genes at all times?
2. Every cell in your body has the same DNA. Explain how a muscle cell and a nerve cell can look and function so differently.
3. What is a transcription factor? How do activators and repressors control gene expression?
4. What is a promoter? What is an enhancer? How far apart can an enhancer be from the gene it regulates?
5. What is a housekeeping gene? Give an example of why some genes must be expressed in every cell.
6. What is an operon? Why are operons common in prokaryotes but not in eukaryotes?
7. Describe how the lac operon works when lactose is absent. What role does the repressor play?
8. Describe how the lac operon works when lactose is present. What happens to the repressor?
9. Why is inducible regulation energy-efficient for the cell?
10. Compare inducible operons (lac) vs. repressible operons (trp). How do they differ in their default state?
11. List the five levels at which gene expression can be regulated (from chromatin to post-translational). At which level is regulation most common?
12. What is alternative splicing? How can one gene produce multiple different proteins?
13. What is miRNA? How does it regulate gene expression after transcription?

14. What are post-translational modifications? Give two examples of how they affect protein function.
15. Define epigenetics. How does it differ from a genetic mutation?
16. What is DNA methylation? Does it typically activate or silence gene expression?
17. What are histones? How does histone acetylation affect gene expression?
18. Explain the difference between euchromatin and heterochromatin. Which allows gene expression?
19. What is X-inactivation? Why does it occur? What is a Barr body?
20. Explain how calico cats demonstrate X-inactivation in action.
21. Explain dosage compensation. Why is it necessary in organisms with XX/XY sex determination?
22. Define point mutation vs. frameshift mutation. Why are frameshift mutations typically more damaging?
23. How can mutations in tumor suppressor genes or proto-oncogenes lead to uncontrolled cell division?
24. Give two examples of how environmental factors (nutrition, stress, toxins, or exercise) can influence gene expression through epigenetic mechanisms.
25. Identical twins share the same DNA. Why might they develop different traits or disease risks as they age?
26. A pregnant individual's diet can influence gene expression patterns in the developing offspring through methylation changes. Explain why folic acid is recommended during pregnancy in light of this.
27. Environmental factors (diet, stress) can modify DNA methylation without changing the sequence. Can these modifications be inherited? Discuss the evidence.

28. A researcher discovers that a tumor suppressor gene in cancer cells has heavy methylation in its promoter region but no mutations in its DNA sequence. Explain how this could contribute to cancer.

29. Compare the lac operon model (prokaryotic) with enhancer/transcription factor regulation (eukaryotic). What is the fundamental principle shared by both systems?

30. If a cell loses the enzyme that performs histone acetylation, predict what would happen to gene expression in that cell. Would it increase or decrease? Why?

31. Diagram the flow: DNA → RNA → Protein. What is the role of DNA polymerase? RNA polymerase? Ribosomes?

32. DNA template strand: 3′-TAC GGG AAA ACT-5′. Transcribe to mRNA (5′→3′) and translate to amino acids using a codon chart.

33. A student says: "Epigenetics proves that Lamarck was right — organisms can pass on acquired characteristics." Evaluate this claim. Is it accurate? What are the limitations?